

XQdrant: In-Database Attribution and Masked-Distance Subspace Search over HNSW

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Abstract

Modern vector databases return opaque scalar similarities over high-dimensional embeddings, leaving practitioners without coordinate-level explanations and without a way to *route* HNSW search using only a dynamic focus subspace. We present **XQdrant**, an extension of the Qdrant engine with two complementary capabilities. First, `with_dims_explained` elevates exact metric score decomposition to a first-class, in-database post-search primitive, avoiding client-side vector egress; on synthetic encoder-width corpora this yields about $2.2\times$ lower attribution latency than post-query extraction at matched Recall@10. Second, we study *masked-distance* HNSW traversal via `focus.masked`, a hybrid layer cutoff (`mask_from_layer`), SIMD gather/repack kernels, an optional verify/overfetch pass, and a weighted score blend (`alpha`). Across SIFT1M and high-dimensional synthetic indices we find that hybrid cutoffs recover navigability (SIFT recall $\sim 0.8\text{--}0.97$ for cutoffs ≥ 1) that naive all-layer masking collapses, that gather kernels win microbenchmarks ($\sim 1.1\text{--}1.8\times$) but do not unlock end-to-end speedups vs. full search, that subspace $\leftrightarrow$ full k -NN coherence predicts when verification can help, and that plain verify at *limit=k* cannot change Recall@ k whereas overfetch+verify recovers full-space recall on coherent data (SIFT) at substantial latency cost. A weighted blend is dominated by a hard layer cutoff. We release an experiment index and reproducible harness so these negative and positive results can be cited independently.

Keywords: vector databases, HNSW, approximate nearest neighbor, explainability, feature attribution, subspace search, masked distance, SIMD

1 Introduction

High-dimensional dense embeddings underpin retrieval-augmented generation (RAG), recommendation, and semantic search [6, 4, 3]. Production engines typically traverse HNSW graphs [1], collapsing each comparison into a single scalar. That design creates two gaps. *Explainability*: practitioners cannot see which coordinates drove a hit without fetching full vectors and re-computing per-dimension terms outside the database. *Subspace routing*: many workloads care about a dynamic focus set $\mathcal{F}$ of coordinates (concept probes, ablations, modality masks), yet stock HNSW always evaluates full D -dimensional distances during graph descent. Preselect-and-rescore on $\mathcal{F}$ improves semantic specificity but cannot reduce traversal work [12].

XQdrant addresses both gaps inside a Qdrant-compatible Rust engine. For attribution, `with_dims_explained` attaches exact top- m coordinate terms after search. For subspace routing, `nearest.focus` with `masked=true` evaluates reduced distances during HNSW, optionally with a hybrid layer cutoff (`mask_from_layer`), gather-based SIMD kernels, a verify/overfetch rescoring mode, and an α -blend of full and masked scores.

Contributions.

- **In-database attribution.** Opt-in exact score decomposition with order-statistic top- m selection; recall-neutral vs. vanilla Qdrant and $\sim 2.2\times$ faster than post-query extraction at encoder widths.
- **Masked-distance HNSW.** Layer-aware dual scorers, API-stable focus fields (`masked`, `mask_from_layer`, `verify`, `alpha`), and gather/repack kernels selectable at runtime.
- **Empirical map of what works.** Hybrid cutoffs restore navigability on SIFT; kernels win microbenches but not wall-clock search; coherence explains high- D plateaus; verify needs overfetch; α -blend is a negative ablation vs. a hard cutoff.
- **Reproducible artifacts.** Scripts under `DB_systems/research/` and a curated experiment index listing canonical CSVs/plots for each claim.

2 Background

2.1 HNSW and Contiguous SIMD

HNSW maintains multi-layer proximity graphs [1]. Queries greedily descend with candidate heap size `ef_search`, then extract top- k at the base layer. Distance kernels assume contiguous `f32` slices (AVX2/NEON), so a non-contiguous focus set either (i) repacks selected coordinates into a scratch buffer or (ii) gathers them in-register/*via* gather loads.

2.2 Score Decomposition

For Dot product, $s(\mathbf{q}, \mathbf{v}) = \sum_i q_i v_i$; Cosine applies the same after ℓ_2 normalization; Euclidean/Manhattan admit analogous additive terms [7, 8]. Attribution is thus exact accounting of the engine’s own metric—not a surrogate XAI model.

2.3 Subspace Coherence

Let $N_k^{\text{full}}(q)$ and $N_k^{\mathcal{F}}(q)$ be exact k -NN sets under the full metric and under $\mathcal{F}$. *Coherence* $|N_k^{\text{full}} \cap N_k^{\mathcal{F}}|/k$ predicts when masked routing can surface full-space neighbors and when a cheap rescore can help.

3 In-Database Attribution

XQdrant injects attribution at the collection coordinator after shard merge, with an early return when `with_dims_explained` is absent. Per explained hit, Algorithm 1 materializes D terms and selects the top- m by absolute contribution in average $\mathcal{O}(D)$ time (`select_nth_unstable`).

Algorithm 1 (In-database top- m attribution).

Input: query $\mathbf{q}$, vector $\mathbf{v}$, D , m , metric δ .

1. Normalize operands if required by δ .
2. For $i = 0, \dots, D - 1$: append $(i, \text{TERM}_i(\mathbf{q}, \mathbf{v}, \delta))$.
3. Return order-statistic top- m by $|t|$, sorted descending.

Figure 1: In-database top- m feature attribution.

`nearest.focus` (without masking) still performs full HNSW preselection then rescoring on $\mathcal{F}$ —useful for interpretability-constrained ranking but not a traversal acceleration (Section 5.2).

4 Masked-Distance HNSW Design

4.1 API Surface

Focus parameters ride inside the existing query body:

```
"focus": {
  "dims": [...], "masked": true,
  "mask_from_layer": 1,    // optional hybrid cutoff
  "verify": true,         // optional full rescore
  "alpha": 0.25           // optional blend on focus layers
}
```

Constraints mirror dense local search: no quantization/turbo, no simultaneous `with_dims_explained` on masked paths, fail-fast if `mask_from_layer/verify/alpha` are set without `masked=true`.

4.2 Option 1: All-Layer Masking

With `masked=true` and cutoff omitted, every HNSW comparison uses only coordinates in $\mathcal{F}$. This is the cheapest semantic change but destroys navigability when the focus metric disagrees with the index geometry.

4.3 Option 2: Hybrid Layer Cutoff

Layer ℓ uses the masked scorer iff $\ell < L$ for cutoff $L=\text{mask_from_layer}$; otherwise full distance. $L=0$ routes to plain nearest (no masking); a large L recovers Option 1. Implementation builds dual raw scorers inside `FilteredScorer` and switches with `set_layer` at layer boundaries—visited-set and heap logic remain unchanged.

4.4 Kernels: Repack vs. Gather (X1)

Masked scoring historically copied focus dims into a scratch buffer then called contiguous SIMD (*repack*). We add an in-place *gather* path (AVX2 `vgatherdps` for Dot/Cosine/Euclid on x86; scalar elsewhere), selected by `XQDRANT_MASKED_KERNEL`.

4.5 Verify and Overfetch (X3)

`verify=true` re-scores returned candidates with pure full distance after traversal (same hook as quantization rescore). At *limit=k* this only reorders a size- k set, so $\text{Recall}@k$ is unchanged. Overfetch (*limit=m·k*, keep top- k) turns masked search into a candidate filter.

4.6 Weighted Blend (Option 4)

On focus layers, $\text{score} = \alpha \cdot s_{\text{full}} + (1 - \alpha) \cdot s_{\text{focus}}$. Coarse hybrid layers stay full-only. Blend alone never reduces full-distance work; it only biases routing.

5 Experimental Evaluation

5.1 Setup

Attribution suite. Synthetic corpora $N=10k$, $Q=500$, $D \in \{768, 1536\}$, seed 42; baselines: vanilla Qdrant, post-query extract+client top- m , and XQdrant. Hardware for those runs: Apple Silicon laptop (suite orchestrator).

Masked-distance suite. HTTP against a release XQdrant binary, typically with `XQDRANT_MASKED_KERNEL=g`. Corpora: SIFT1M ($D=128$) and synthetic $N=50k$ at $D \in \{768, 1536\}$, $Q=500$, $k=10$, `ef_search=128`. Canonical folders are listed in `DB.systems/experiments/README.md`; figures below are copied into `paper/figures/`.

5.2 Attribution and Focus Rescoring

Figure 2 shows that enabling `with_dims_explained` preserves the latency–recall frontier vs. vanilla Qdrant at $D=768$ (Recall@10 gap < 0.01). Figure 3: in-database attribution holds ~ 3.5 ms p50 across $m \in \{1, \dots, 50\}$ while post-query extraction sits near 7.8 ms ($\approx 2.2\times$). Figure 4: preselect-and-rescore focus yields speedup < 1 (0.84–0.87), motivating masked traversal rather than rescoring alone.

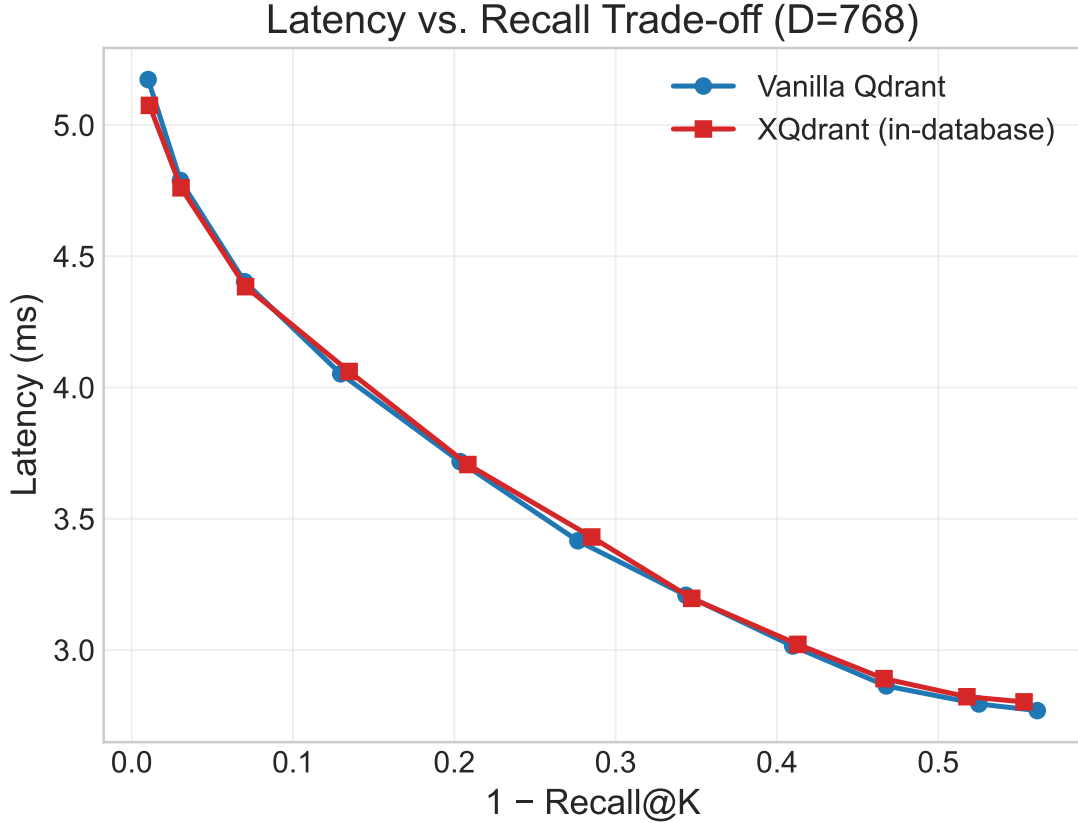


Figure 2: Attribution suite: latency–recall at $D=768$ with `with_dims_explained` ($m=10$).

5.3 Option 1: Naive Masking Collapses Full Recall

Figure 5: on SIFT1M, all-layer masked search vs. *full* ground truth yields Recall@10 $\approx 0.006/0.079/0.257/0.464$ at ratios 0.1/0.25/0.5/0.75/1.0.

5.4 Option 2: Hybrid Cutoff Recovers Navigability

Figure 6: with gather kernels, SIFT hybrid cutoffs $L \geq 1$ recover ~ 0.79 – 0.98 recall (script subspace/hybrid GT), while $L=0$ (plain nearest vs. that GT) tracks low coherence. High- D synthetic hybrids lift collapsed cells (e.g., $D=768$, ratio 0.25: $\sim 0.01 \rightarrow \sim 0.20$) but plateau near 0.15–0.22. **End-to-end speedup vs. full search remains < 1** in every gathered cell (Figure 7).

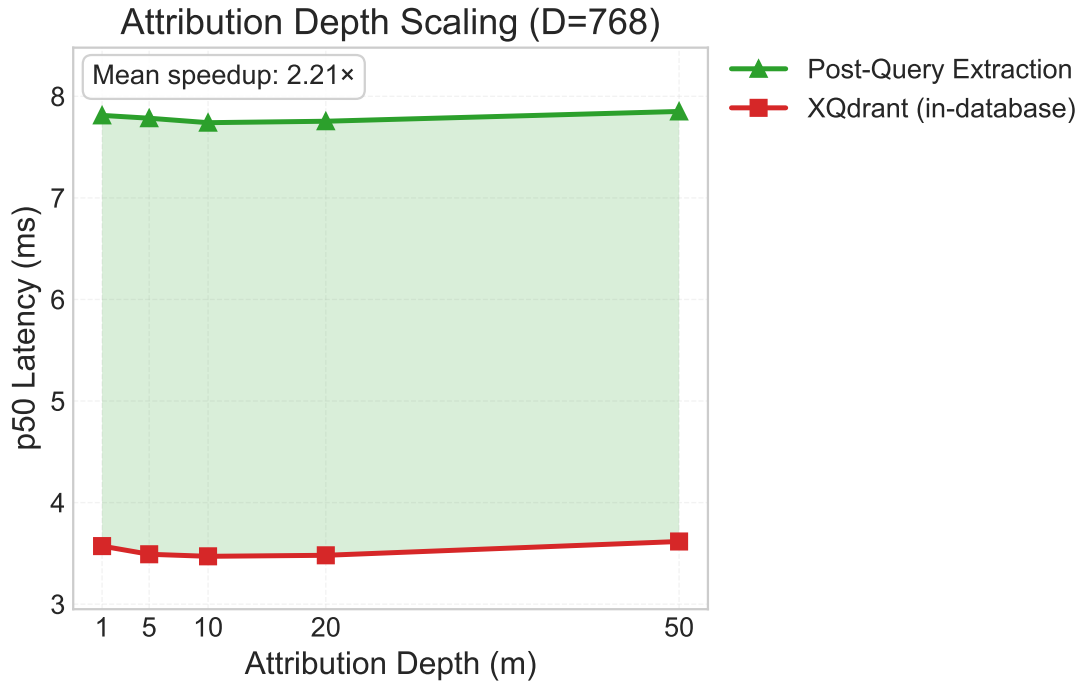


Figure 3: Attribution depth: in-DB vs. post-query extraction ($D=768$).

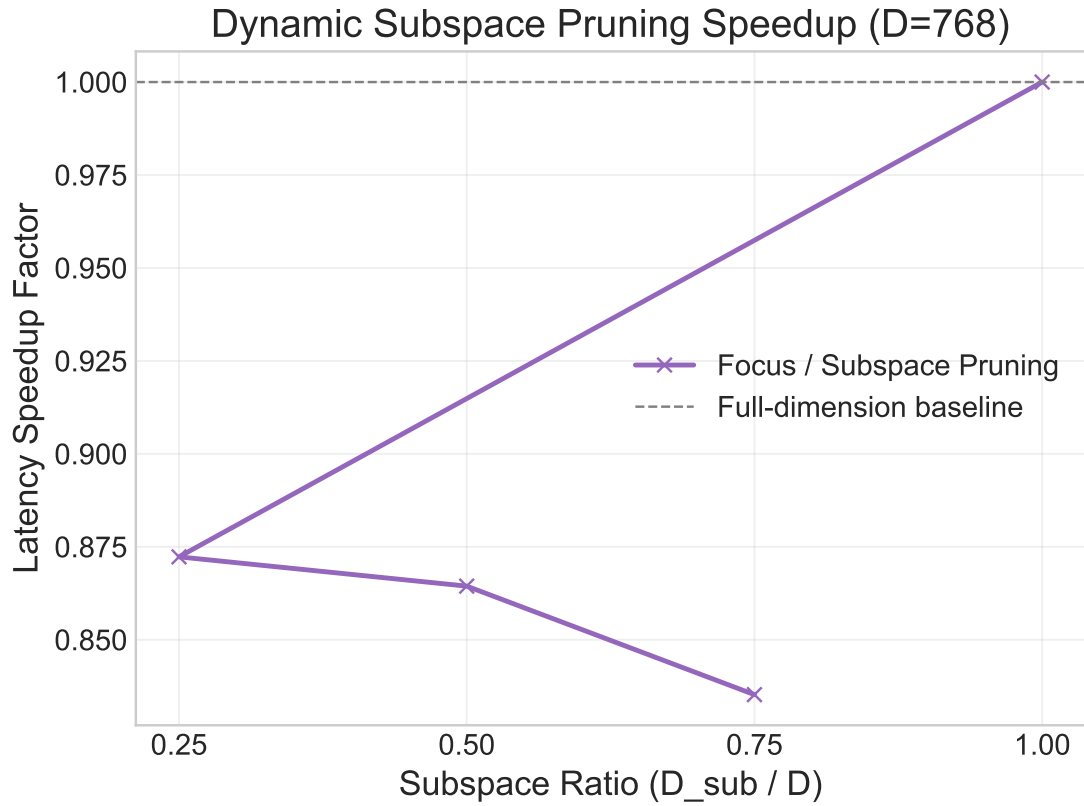


Figure 4: Focus *rescoring* speedup $T_{\text{full}}/T_{\text{sub}}$ ($D=768$): values < 1 show rescoring does not accelerate search.

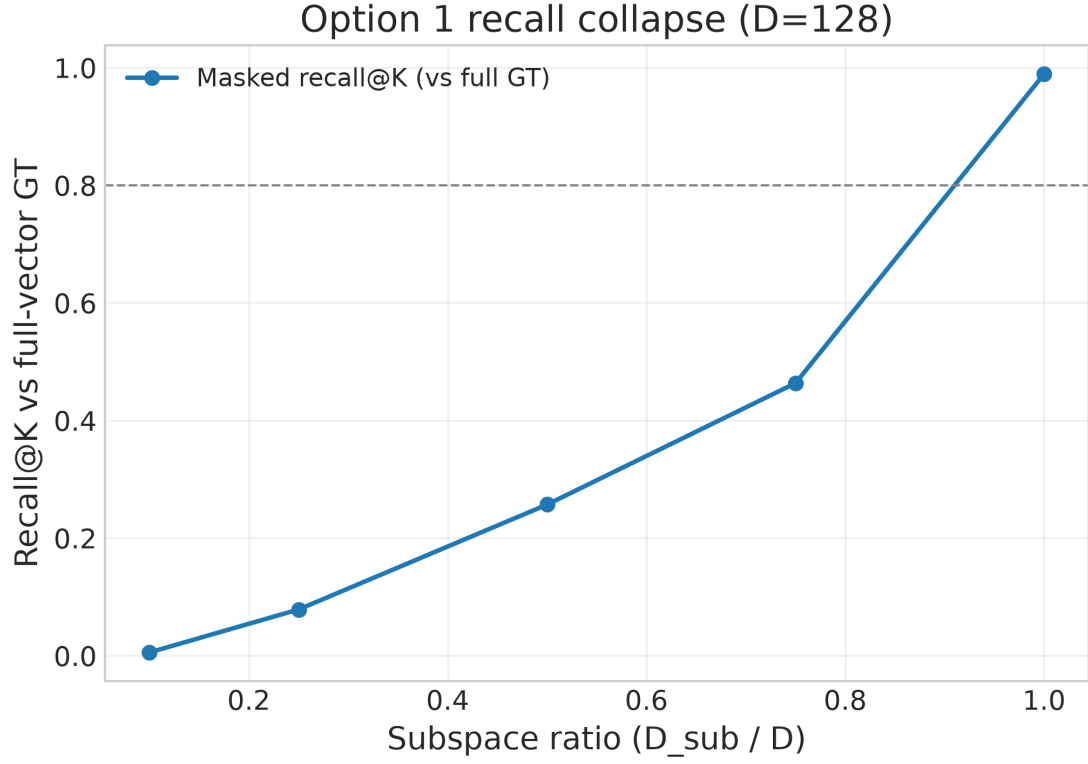


Figure 5: Option 1 on SIFT1M: Recall@10 vs. full-space GT as D_{sub}/D shrinks.

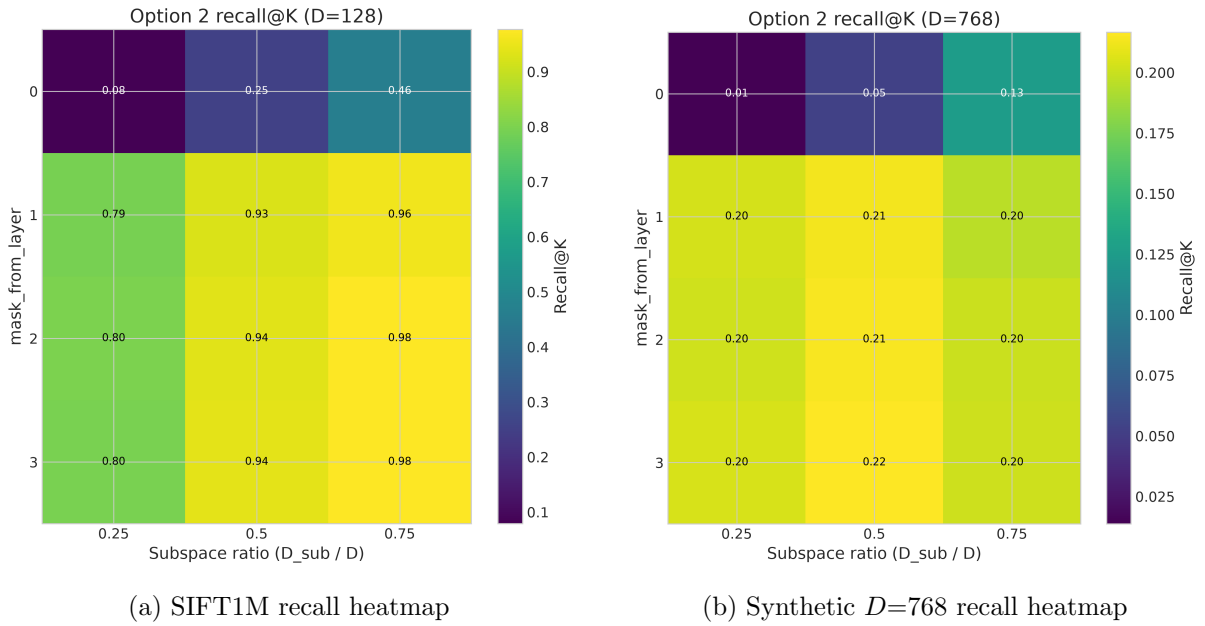


Figure 6: Option 2 ($\text{mask_from_layer} \times \text{ratio}$) with gather kernel.

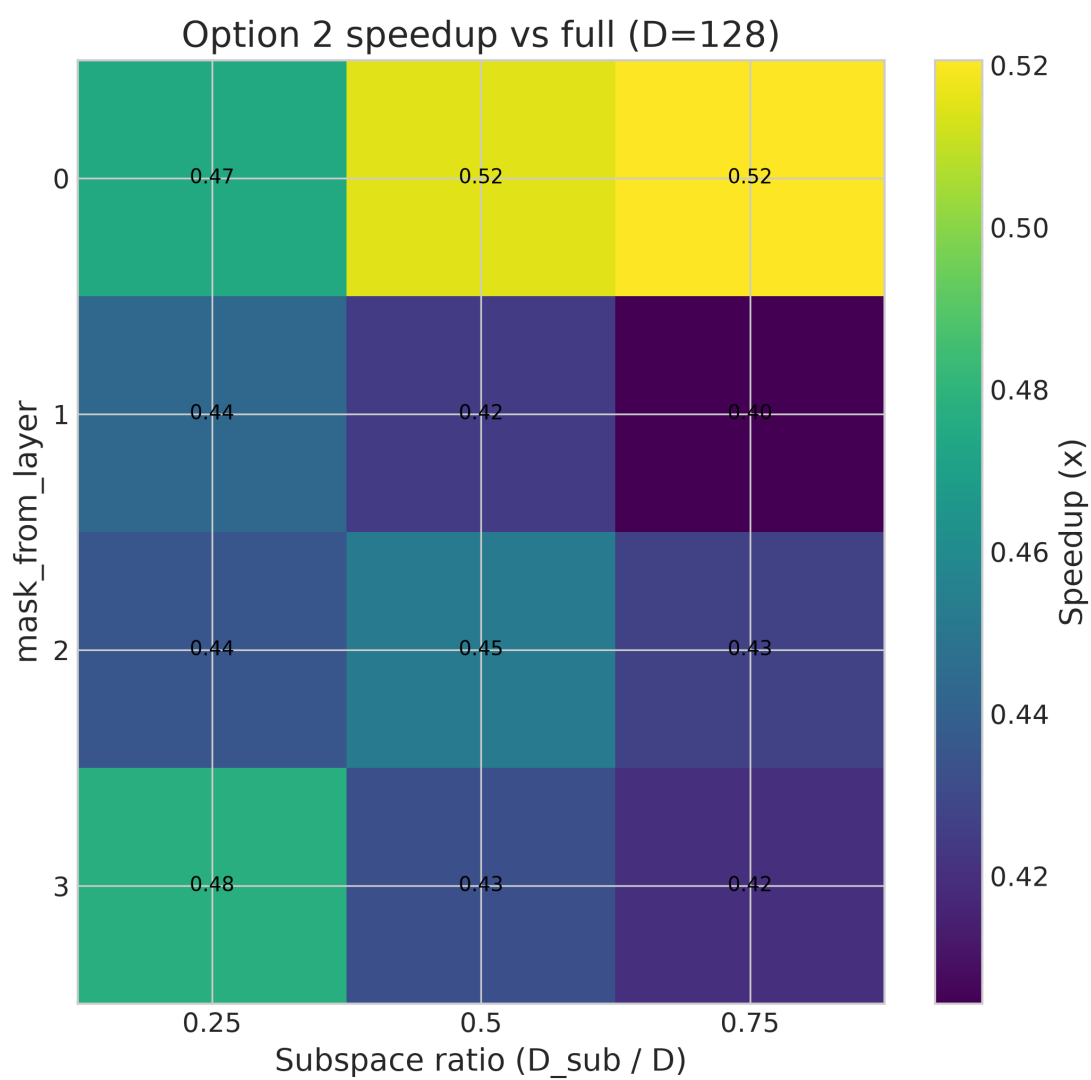


Figure 7: Option 2 SIFT speedup vs. full search: no cell exceeds $1\times$.

5.5 X1: Gather Wins Microbenchmarks, Not Wall-Clock

Figure 8: gather is $\sim 1.24\times/1.14\times/1.42\times/1.77\times$ faster than repack at $D_{\text{sub}} \in \{32, 128, 384, 768\}$ (ns/op). Re-running Option 2 with gather leaves recall intact but still shows **zero** end-to-end cells with speedup > 1 .

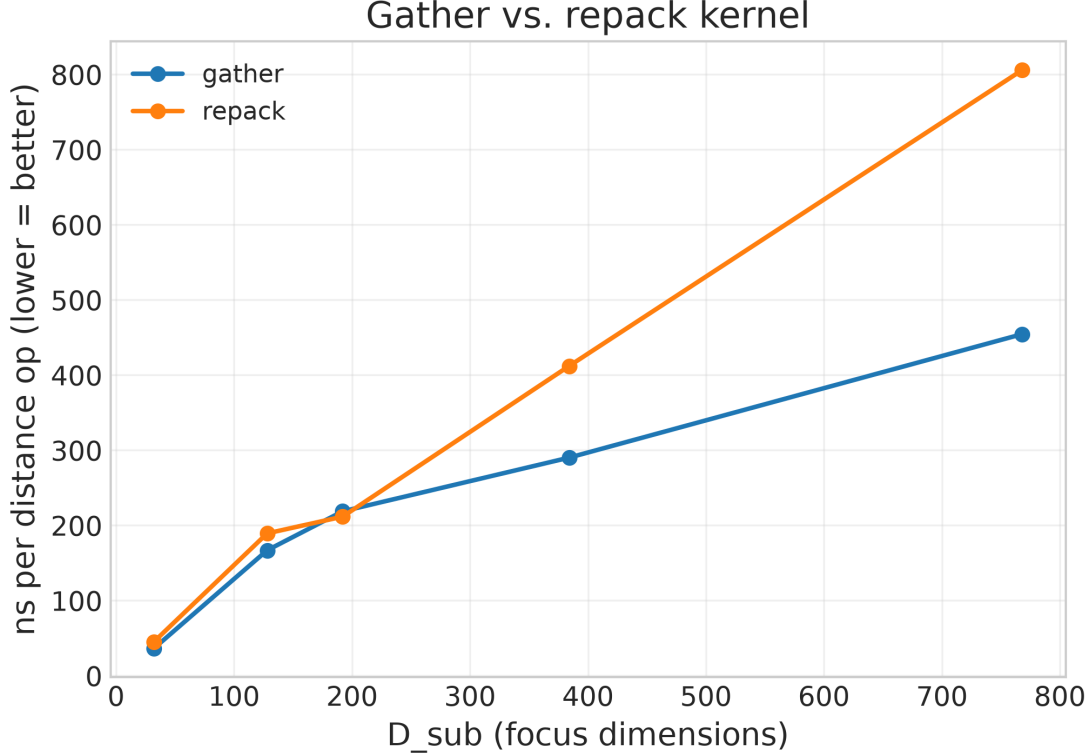


Figure 8: X1: masked Dot kernel throughput (repack vs. gather).

5.6 X2: Coherence Predicts High- D Failure

Figure 9: overlap@10 grows with ratio but stays low on synthetic $D=768$ ($\sim 0.007 \rightarrow 0.30$ from ratio $0.1 \rightarrow 0.75$); SIFT is somewhat higher ($\sim 0.01 \rightarrow 0.46$). This explains why hybrid navigability does not imply full-space recall recovery.

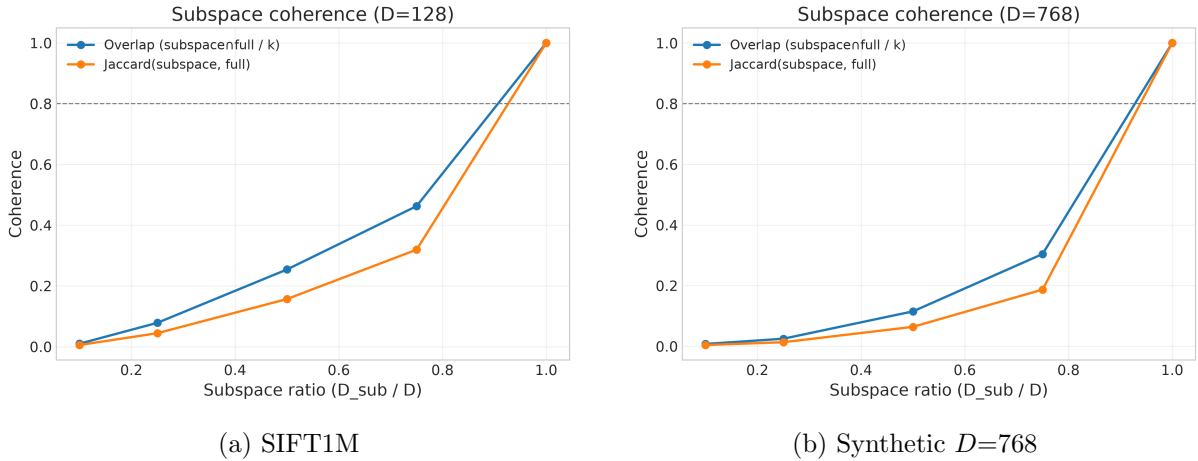


Figure 9: X2: subspace $\leftrightarrow$ full exact top-10 overlap vs. ratio.

5.7 X3: Verify Needs Overfetch; Costly When It Helps

Evaluating against **full-space** GT (corrected harness): at overfetch $m=1$, masked and verify Recall@10 are identical (same k IDs). At $m=8$ on SIFT with hybrid $L=1$, full recall rises e.g. $0.26 \rightarrow 0.66$ (ratio 0.5) and $0.45 \rightarrow 0.89$ (ratio 0.75), but p50 jumps by $\sim +45$ ms ($\sim 1.5 \rightarrow \sim 46$ ms). High- D synthetic stays $\leq \sim 0.19$ even with $m=8$ (Figure 10).

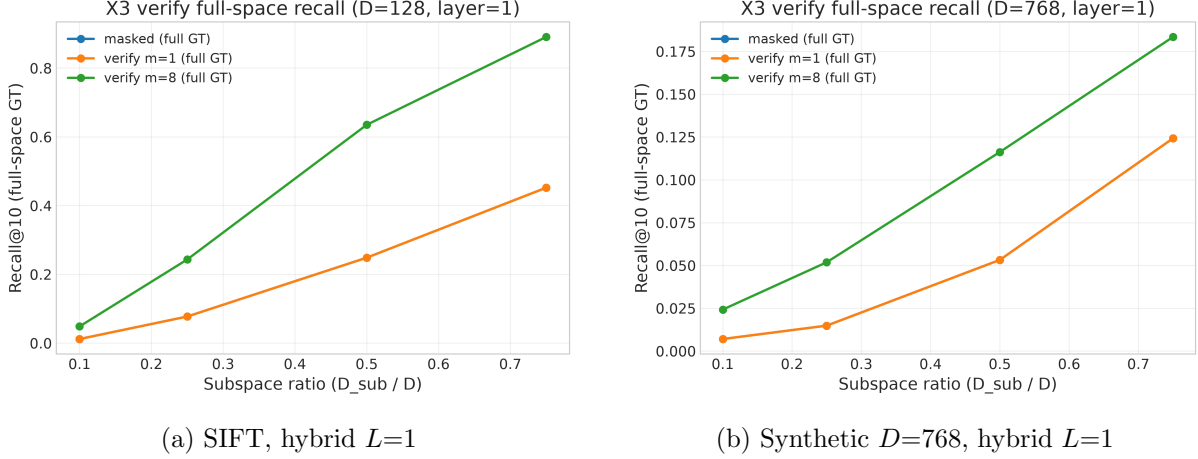


Figure 10: X3: full-space Recall@10 for masked vs. verify at overfetch $m \in \{1, 8\}$.

5.8 Option 4: Blend Is Dominated by $\alpha=0$

Figure 11: at ratio 0.25, SIFT hybrid recall is best at $\alpha=0$ (~ 0.84 for $L \geq 1$) and collapses monotonically toward the coherence floor as $\alpha \rightarrow 1$. Hard `mask_from_layer` without blend is strictly preferable.

5.9 Summary Table

Table 1: Keep / demote map for the masked-distance study (canonical runs).

Knob	Verdict	Evidence
Attribution (in-DB)	Keep	$\sim 2.2\times$ vs. post-query; recall-neutral
Focus rescoring only	Not accel.	Speedup < 1 (Fig. 4)
Option 1 all-mask	Baseline	Full-GT collapse on SIFT (Fig. 5)
Option 2 hybrid	Keep	Navigability recovery; e2e speedup still < 1
X1 gather kernel	Micro keep	$1.1\text{--}1.8\times$ ns/op; e2e still no win
X2 coherence	Keep diag.	Predicts high- D / X3 limits
X3 verify $m=1$	API only	Recall@ k unchanged
X3 + overfetch	Conditional	Helps SIFT; $\sim +45$ ms; fails high- D
Option 4 α	Skip enh.	Best at $\alpha=0$ (= Option 2)

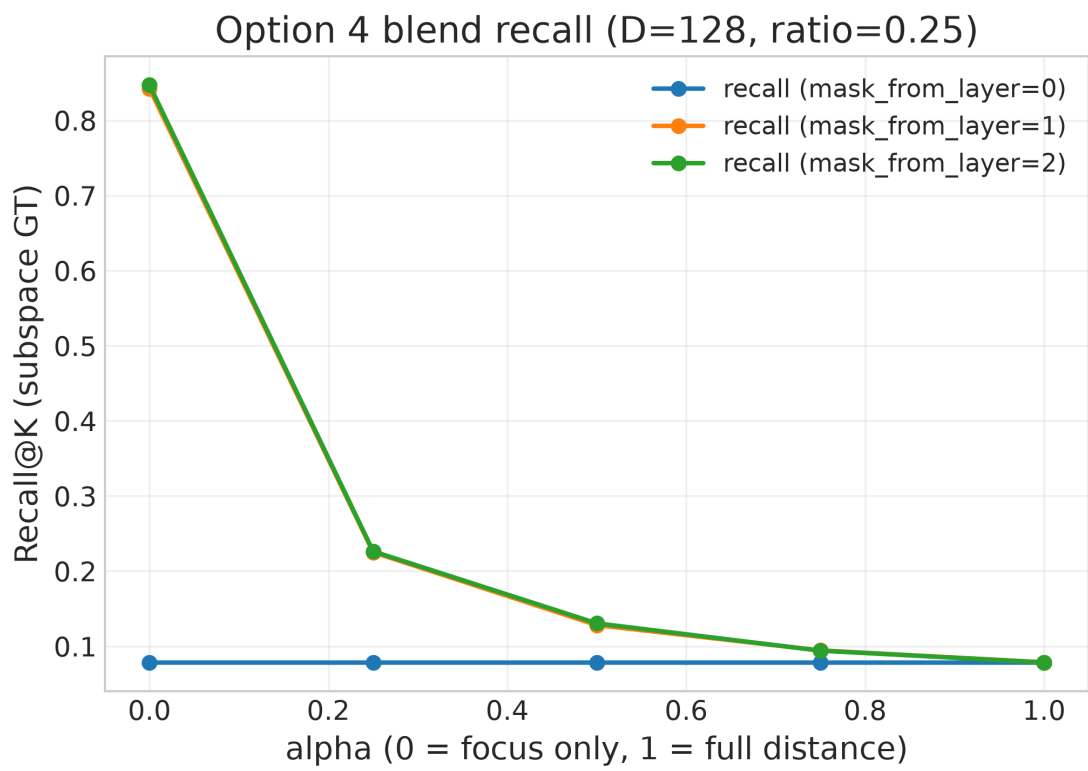


Figure 11: Option 4 on SIFT: recall vs. α for cutoffs $\{0, 1, 2\}$ (ratio 0.25, subspace GT).

6 Related Work

Vector databases. Milvus [3], Pinecone [5], and Qdrant [4] expose ANN with filters and quantization but not native per-coordinate decompositions or layer-aware masked metrics.

ANN libraries. FAISS [2] and ScaNN [9] optimize large-scale search; explainability and dynamic subspace routing remain post-processing.

Explainable AI. SHAP/LIME [7, 8] approximate model behavior. We instead exploit additive metric structure for deterministic attributions aligned with ranking, closer to OLAP rollups [10].

Dimension reduction / projections. Static projections or multi-index layouts differ from *query-time* focus sets over an existing full-dim HNSW. Projected auxiliary indexes (Option 3) remain future work when focus palettes are small and known a priori.

7 Discussion

What to ship. In-database attribution and hybrid `mask_from_layer` are the positive systems results. Gather kernels are worth keeping as an implementation detail. Verify is useful as an optional correctness mode *with* overfetch on coherent data; α -blend should not be marketed as improving Option 2.

Why no e2e speedup. Even with faster masked compares, HNSW still touches full vectors for storage, graph structure, and (on hybrid coarse layers) full SIMD. I/O and candidate management dominate the microbenchmarked arithmetic.

Limitations. High- D Option 1 lacks a dedicated archive folder (high- D collapse is evidenced via Option 2/X2). We did not instrument visited-set divergence (X4) or build projected indexes (Option 3). Attribution suite hardware differs from the Linux masked-suite host; absolute milliseconds are not cross-compared across suites.

8 Conclusion

XQdrant shows that exact feature attribution can live inside a Rust vector engine with negligible quality loss, and that masked-distance HNSW is a viable *navigability* tool when coarse layers keep full distance—but not yet a latency win, and not a substitute for coherent subspaces when the goal is full-space recall. Overfetch+verify and α -blending clarify the design space: the former can recover recall when candidates are in range; the latter is dominated by a hard layer cutoff. These measured positives and negatives, backed by a public experiment index, form a concrete basis for subspace-aware vector search in production engines.

Acknowledgments

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